



• IIC 3.0 MUJ • THEME: EDTECH • PS-08 ACADEMIA-INDUSTRY COLLABORATION

ProofBridge

Cryptographic Skill Verification & Talent Intelligence Network

Evidence-Backed Portfolios • W3C Standard Proofs • Shortest-Path Employability Engine

"Stop trusting what students claim. Mathematically verify what they can build."

PROBLEM STATEMENT

PS-08 · EdTech

Portal for Academia-Industry collaboration for skill mapping, internships & placements.

TEAM NAME

Outliers

Bridging universities, student engineering artifacts, and enterprise recruitment.

TEAM MEMBERS & ROLES

- **Akshar Miyani**
Lead Architect & Cryptography
- **Lakshita Sharma**
Academic Rubrics & Dataform

PROOFBRIDGE • IIC 3.0 MUJ PITCH • FINAL ROUND

01 · THE CRISIS IN RECRUITMENT

Resumes Are Broken. Keyword

Generative AI has flooded recruitment portals with fabricated, keyword-stuffed CVs. Degrees signify not production software competence.

THE TRADITIONAL PORTAL ILLUSION



Student Claims "Python 90%"



AI-Generated PDF Resume



AI

78% RESUME INFLATION

Unverified Claims & Ghost Skills

Over 78% of student engineering resumes contain exaggerated skills. Writing "Expert in React & SQL" costs nothing, forcing recruiters into endless rounds of elementary verification.

Zero verification at the point of submission.

64% DEGREE DETACHMENT

Grades Don't Reflect Tool Competence

A university transcript awards an "A" in Database Systems, but cannot verify if the candidate can handle messy production data parsing, LAG window functions, or write defensive queries.

Curriculum gap remains hidden from professors.

Claiming a skill ≠ Proving you can

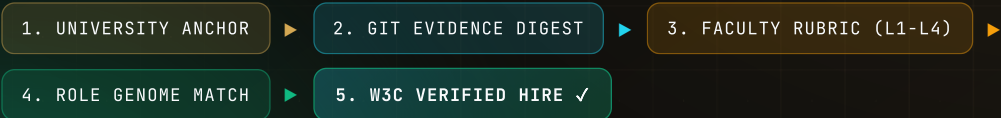
THE MARKET GAP IN PS-08

Existing portals (LinkedIn, Internshala, Unstop) treat skills as **self-typed text strings**.

Companies don't want "keyword matches"; they need **mathematically verified proof** that a student has handled messy datasets, written automated tests, and adhered to software quality benchmarks.

OUR SOLUTION – THE CRYPTOGRAPHIC EVIDENCE CHAIN

We forge the missing connection between education & work:



Zero LLM hallucination • 100% human-verified provenance • W3C Standard.

ECOSYSTEM ST

Who Pr

**St**
SKID

Ditch unreal
Twin, see ex
micro-missi

Benefit: Aut

**En**
BLI

Publish requ
repositories
paying ATS

Benefit: 80%

ProofBridge: The Cryptographic

An evidence-based talent intelligence network that replaces unverified resumes with mathematical proof anchored by accredited universities.

PILLAR 01

Skill Twin
Continuous digital twin of verified student competence derived from real code repositories, not self-reported CV text.

- Frozen Git Revisions (SHA-256)

PILLAR 02

Role Genome
Job requisitions converted into multi-factor capability vectors with exact rubric levels (L1-L4) and percentage weights.

- Zero keyword ambiguity

PILLAR 03

Bridge Me E
Algorithmic pathfinding missions to bridge employer needs (match).

- Shortest-path em

OPERATIONAL WORKFLOW

How ProofBridge Operates in Real Time

1 CHALLENGE SANDBOX

Candidate Solves Challenge
Student writes SQL/code against uncurated enterprise anomalies. Validates results in interactive test runner.

2 ATOMIC CODE FREEZE

Revision Lock & Audit
PostgreSQL creates an immutable frozen revision with SHA-256 hash. Student writes contribution defense & AI disclosure.

3 FACULTY A

Standardized R
Accredited faculty i (L1-L4). Issues insti

Real Code Repositories as Group

ProofBridge parses GitHub commits, AST structures, and test suites to construct an evidence-backed mandatory author contribution defense.

EVIDENCE-BACKED SKILL GRAPH EXAMPLE

Akshar Miyani · Candidate Profile #001

C++ Systems Engineering87% VERIFIED

DEMONSTRATED EVIDENCE (FROM 4 GITHUB REPOSITORIES):

- ✓ 4 C++ repositories linked & analyzed
- ✓ Extensively used STL containers, smart pointers & memory guards
- ✓ Built full-featured CLI multi-module architecture (2,840 lines)
- ✓ Direct OS file-system and low-level POSIX interactions

IDENTIFIED WEAK AREAS & DEFICITS:

- ✗ Limited automated unit testing (41% coverage)
- ✗ No multithreaded concurrency or mutex synchronization evidence

SQL & Analytics

Level 3 Proficient

LAG window functions, COALESCE guards

API Integration

Level 2 Developing

FastAPI, exponential backoff retries

Mandarin

AUTHOR

Candidates must demonstrate COALESCE and LAG grade this directly

AI & Tooling

NO SHAM

We don't ban AI; we require Syntax, or (3) Built

Frozen

TAMPER -

Once submitted, candidate nor fa

PROOFBRIDGE · GITHUB GROUND TRUTH & EVIDENCE · IIC 3.0 MUJ

file:///D:/IIC3.0/ProofBridge/PPT/slides/05-features.html

1/1

Production Architecture & Dete

Built on open W3C standards with zero recurring API costs. Completely deterministic scoring elim

CLIENT & WORKSPACE

NEXT.JS 14

Reactive Enterprise Frontend

Fully responsive workspace providing isolated environments for Students, Faculty Reviewers, Employers, and University Deans.

Framework

Next.js 14 (App Router)

Language

TypeScript (Strict Mode)

Styling

Tailwind CSS + ThemeToggle

Motion

Framer Motion micro-interactions

Zero layout shifts · Client-side persona isolation

PERSISTENCE & API

PostgreSQL Immutable Store

Relational engine managing immutable code revisions, crypto hashes, standardized rubrics, and dynamic live state.

Database

PostgreSQL with Con

Revision History

Immutable App

API Layer

RESTful Route Handle

State Engine

Atomic Syn

Atomically signed transactions · No race conditions

Curriculum Intelligence & Blind T

Connecting university academic planning with real-time employer demand, while removing pedig

UNIVERSITY LEADERSHIP PERSPECTIVE

Curriculum Gap Radar · Manipal University Jaipur

Instead of discovering placement failures at the end of the year, deans view real-time market capability deficits across the student body.

Cohort Analysis: Final-Year Computing (620 Students)

Live Sync

142

Industry-Ready Now

213

Near-Ready (1 Challenge Away)

265

Needs Core Scaffolding

TOP CURRICULUM GAPS DEMANDED BY INDUSTRY PARTNERS:

Docker & Containerization

49% Gap Deficit

Cloud Deployment & CI/CD

44% Gap Deficit

Automated Unit & Integration Testing

38% Gap Deficit

Actionable Outcome: Department head schedules 2 weekend bootcamps, closing 70% of gap before hiring season.

EMPLOYER TA

Blind Vec

Employers sea
candidate is m

Blind Cand

- Target Ro
- SQL (Leve
- Code Rev
- Tooling:

1-Click Sho

Eliminates T

Students from

Direct Pipeli

Every faculty s

Economic Feasibility & Accreditation

Engineered for zero-cost operation at scale, complete compliance with national educational accreditation standards, and a quantifiable ROI.

ECONOMIC VIABILITY

Zero-Cost Scaling

Unlike platforms that burn thousands of dollars on monthly LLM tokens or proprietary verification APIs, ProofBridge relies on deterministic code execution and faculty scoring.

- ✓ Zero OpenAI / Proprietary API dependencies
- ✓ Self-hosted PostgreSQL & Next.js runtime
- ✓ Zero vendor lock-in with open W3C standards

Marginal Cost Per Student Verification: = ₹0.00

INSTITUTIONAL ALIGNMENT

Accreditation Standards

ProofBridge directly provides the verifiable data required for university accreditation frameworks in India and international standards.

- ✓ NAAC Criterion 2 & 5: Student progression & placement
- ✓ NBA OBE: Attainment of Program Outcomes (POs & PSOs)
- ✓ NEP 2020: Modular, micro-credential skill pathway

Automated Export for NAAC/NBA Audits

Team Outliers - The Future of Pro

Four builders united to solve the academia-industry divide through unalterable software proof.

AM

Akshar Miyani
LEAD ARCHITECT & CRYPTOGRAPHY

Engineered the coverage-v1 matching algorithm, PostgreSQL immutable revision schema, and SHA-256 digest architecture.

Core Architecture • Full-Stack

LM

Lubhanshi Mathur
SYSTEM ENGINEERING & UX

Designed the multi-stakeholder persona workflows, Bridge Me interactive simulator UI, and responsive design system.

Frontend Experience • Human Factors

LS

Lakshita Sharma
ACADEMIC RUBRICS & OBE

Mapped standardized 4-tier rubrics to NAAC A+ and NBA Criterion 2.6 Outcome-Based Education frameworks.

Institutional Rubrics • Accreditation

MF

Mohammad Faizan
BACKEND PIPELINES & TESTING

Constructed automated SQL evaluation sandbox test suites, API routing handlers, and state management synchronization.

Backend Services • CI/CD Pipelines

EVOLUTIO

Strate

PHASE

Full W

Working

queue l

PHASE

GitHu

Direct C

across

PHASE

Enterp

Direct o

instituti